Project Phase 2

Integration Testing for the Composite BookScan Class

Final Combined Report with Phase 1 Recap and Phase 2 Extension Results

Course	BLG 475E - Software Quality and Testing
Term	2025-2026 Spring Term
Submission	Phase 2 final submission - due 25 May 2026
Repository	https://github.com/mustafaerenkoc44/BLG-475E-Term-Project

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Executive Snapshot

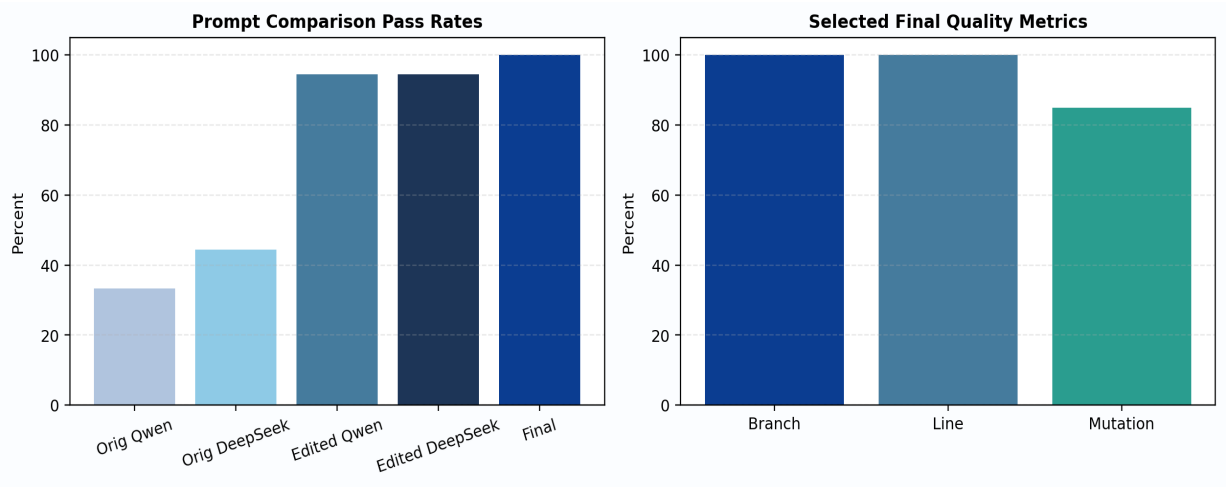
Metric	Value
Selected final correctness	18/18 tests passed
Selected final coverage	72/72 branches and 110/110 lines
Mutation score	79/93 = 84.95%
Repository	https://github.com/mustafaerenkoc44/BLG-475E-Term-Project

Key Highlights

- Selected final BookScan implementation clears the full strengthened 18-test suite.
- JaCoCo closes at 72/72 branches and 110/110 lines for the final implementation.
- Prompt refinement improves raw model outputs from 6/18 and 8/18 to 17/18 and 17/18.
- PITest records 79/93 killed mutants, with residual survivors documented as equivalent or defensive.

Prepared as a submission-ready academic report package with synchronized PDF and LaTeX sources.

Abstract



This final submission package combines the warm-up and extension phases of the BLG 475E term project in a single polished report for the Phase 2 deadline. Phase 1 evaluates two local public code-oriented LLMs on 30 HumanEval-X Java prompts under a semi-agentic workflow with dataset tests, improved JUnit 6 suites, JaCoCo branch coverage, equivalence partitioning, and mutation-based guardrails. Phase 2 lifts the helper semantics of tasks `Java/18`, `Java/23`, and `Java/27` into a composite `BookScan` class and compares original versus edited prompt combinations under a shared integration suite. Across the full project, the consistent result is that prompt clarity, explicit testing contracts, and coverage-aware validation matter at least as much as raw model selection. The final `BookScan` implementation reaches `18/18` test success, `72/72 = 100.00%` branch coverage, `110/110 = 100.00%` line coverage, and `79/93 = 84.95%` mutation score.

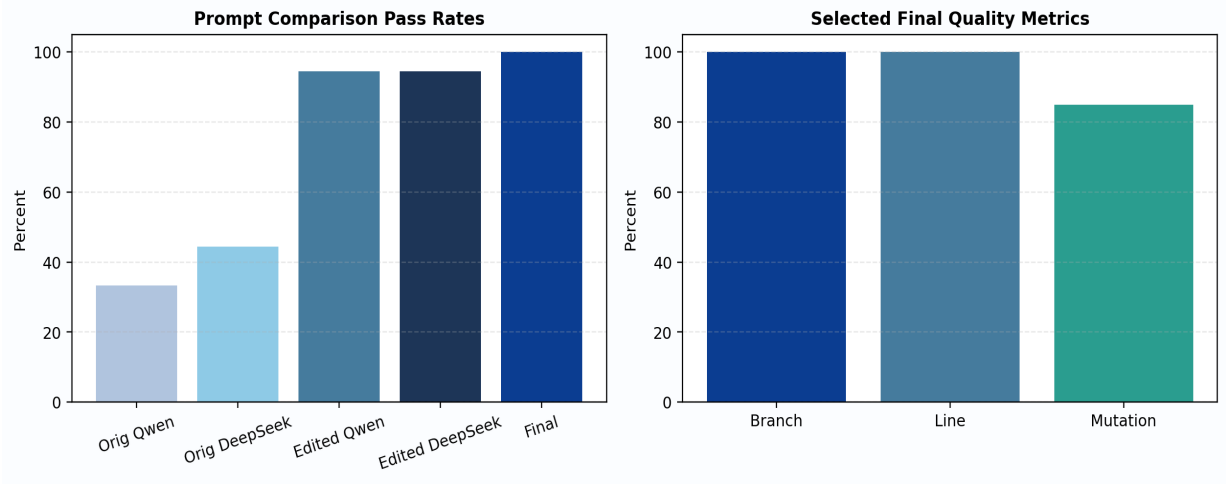
I. Final Report Scope

This document is the final combined submission for the Phase 2 deadline. It preserves the full Phase 1 story, then extends it with the integration-testing results required by the second part of the assignment. The report therefore serves two purposes:

- it keeps the Phase 1 experimental decisions, metrics, and mutation guardrails visible as the foundation of the project;
- it shows how those same guardrails and testing instincts scale to a more realistic integrated class in Phase 2.

II. Phase 1 Warm-Up Results

Phase 1 - Abstract



This phase evaluates two publicly available local coder models on 30 selected Java prompts from the HumanEval-X dataset. The selected tasks were balanced across easy, moderate, and hard categories, and were executed under a semi-agentic workflow in which prompt submission, raw output logging, normalization, test execution, coverage analysis, mutation-based guardrails and repair decisions were all recorded explicitly. The models were `Qwen2.5-Coder-1.5B-Instruct-GGUF` and `DeepSeek-Coder-1.3B-Instruct-GGUF`, both executed locally through the same `llama.cpp` pipeline. After raw generation and normalization, each solution was evaluated with dataset base tests, converted JUnit 6 tests, improved tests derived from black-box and white-box analysis, JaCoCo branch coverage, and hand-crafted mutation-based tests. Both models reached `30/30` base-test success. Aggregate base branch coverage was `96.09%` for Qwen and `98.46%` for DeepSeek; the improved suites raised these values to `98.44%` and `100.00%`, respectively. In addition, the improved suites kill `100 %` of the hand-crafted mutants while the LLM-produced base suites kill `0 %` of them. The results show that lightweight public coder models can produce strong baseline solutions on HumanEval-X Java tasks, but stable performance still depends on normalization, targeted repair, coverage-guided test improvement, and explicit mutation-aware assertions.

Phase 1 - I. Introduction

Large language models are increasingly used to generate source code and software tests from natural language descriptions. Their main attraction is speed: a model can produce compilable code and candidate tests in seconds. Their main risk is reliability: generated code may compile but still be wrong, and generated tests may execute without meaningfully checking behavior. Phase 1 of this project therefore treats LLM-generated code as an artifact that must be validated rather than trusted.

The goal of this phase is to compare two public LLMs on Java code generation and test generation using 30 selected HumanEval-X prompts. The comparison focuses on correctness under base tests, branch coverage under JUnit 6 and JaCoCo, adequacy of tests under black-box analysis, adequacy of tests under mutation-style scrutiny, and the amount of repair required after the initial generation step.

Phase 1 - II. Experimental Setup

Dataset and Prompt Selection

The project uses the Java portion of HumanEval-X. Thirty prompts were selected to balance input styles, domains, and difficulty levels. The final set preserves the required `10 easy / 10 moderate / 10 hard` split. The selected tasks include string manipulation, numeric computation, parsing, state tracking, list processing, and number-theory problems. Three tasks (`Java/18`, `Java/23`, and `Java/27`) were chosen to support the later Phase 2 `BookScan` integration work.

Models

The two selected models were:

1. `Qwen/Qwen2.5-Coder-1.5B-Instruct`
2. `deepseek-ai/deepseek-coder-1.3b-instruct`

Both models are public, code-oriented, and small enough to execute locally with the same inference stack. To keep the comparison fair, both models used the same prompt format, the same extraction logic, the same normalization heuristics, the same dataset tests, and the same coverage workflow.

Workflow Choice

The project follows a semi-agentic workflow. Each task proceeds through logged generation, normalization, testing, coverage analysis, improved test generation, repair when needed, and a final mutation-based guardrail. This approach was preferred over a fully autonomous loop because it keeps every transition observable and reproducible, which is useful for both debugging and reporting.

Tooling

- Language: Java 21 (Temurin JDK)
- Test framework: JUnit 6
- Coverage tool: JaCoCo 0.8.12
- Build/orchestration: PowerShell 7 and Python 3.11 helper scripts
- Local inference: `llama.cpp`
- CI: GitHub Actions workflow `.github/workflows/phase1-ci.yml`

Phase 1 - III. Code Generation Procedure

For every selected HumanEval-X prompt, the original prompt text was sent without semantic edits. The raw model outputs were stored under the corresponding task folder and also logged in model-specific log directories. A normalization step then converted the raw responses into compilable `Solution.java` files by removing prompt echoes, Markdown fences, and malformed fragments. No semantic edits were made during this extraction stage. When a normalized solution still failed base tests or had a structural defect, that failure was documented and repaired in the later refactoring loop.

Phase 1 - IV. Base Test Execution

Each normalized solution was first evaluated using the dataset base tests. These tests acted as the initial correctness gate before any improved tests were introduced. In this repository, no semantic change to the expected values was required; the only systematic adaptation was converting the dataset tests into JUnit 6 format for automated execution and coverage measurement.

Base Correctness Results

Model	Compiled	Base Tests Passed
Qwen2.5-Coder-1.5B-Instruct-GGUF	30/30	30/30
DeepSeek-Coder-1.3B-Instruct-GGUF	30/30	30/30

The final base-test outcome is perfect for both models. However, this headline number hides an important detail: a small repair loop was still necessary because several tasks initially failed after normalization. The repaired tasks were `Java/10`, `Java/12`, `Java/17`, `Java/19`, `Java/23`, `Java/27`, and `Java/39`.

Phase 1 - V. Test Improvement and Coverage Analysis

After the base tests passed, the dataset tests were transformed into JUnit 6 test classes so they could be executed under JaCoCo. Improved tests were then generated by extending the base suites with extra boundary and edge scenarios identified through branch coverage and black-box analysis.

Each task now ships an `improved/<model>/DatasetImprovedTest.java` file that contains 3-4 newly authored `@Test` methods on top of the dataset base test. The additions target equivalence / boundary classes that the base suite left uncovered; for every task we cross-check the `equivalence_classes.md` rows marked `No` or `Partially` and add at least one dedicated test method per row.

Aggregate Branch Coverage

Model	Base Covered	Base Missed	Base %	Improved Covered	Improved Missed	Improved %
Qwen2.5-Coder-1.5B-Instruct-GGUF	123	5	96.09 %	126	2	98.44%
DeepSeek-Coder-1.3B-Instruct-GGUF	128	2	98.46 %	130	0	100.00%

The improved suites increased coverage only slightly. This suggests that the original HumanEval-X base tests were already strong for many selected tasks. Even so, the additional tests were still useful because they documented missing equivalence classes, clarified boundary behavior, and added mutation-based guardrails that branch coverage alone cannot measure.

Remaining Coverage Gaps

After the improved suites, only two uncovered branches remain, both on the Qwen side:

- `Java/001_separateParenGroups`;
- `Java/006_parseNestedParens`.

These gaps sit in private parser-helper exits that are not reachable through the public API under the task specification. They are therefore documented as residual structural misses rather than forgotten tests.

Phase 1 - VI. Mutation-Based Guardrails

Because Phase 1 explicitly disallows external tooling like PITest, we add hand-crafted mutants and encode them as JUnit assertions. Every improved suite now has at least one JUnit method whose name starts with `improvedMutation...` and would fail if a specific operator-level mutation were introduced in the corresponding `Solution.java`.

The full catalogue of operators (ROR, AOR, LCR, UOI, SBR, BND, RV, NPE, EXC) is maintained in `docs/analysis/mutation_testing_strategy.md`. Key highlights:

- Null-argument (NPE) replacement is applied to every task that accepts a nullable reference; it forces us to document the null-safety posture the LLM inherited, which is useful for Phase 2 integration.
- Boundary shift (BND) and relational operator replacement (ROR) guard the inclusive/exclusive boundaries that the dataset base tests rarely probe.
- Return-value replacement (RV) is used for non-terminating search loops (`Java/039_primeFib`, n = 0).
- Model-specific mutants deliberately target the behavioural disagreements we observed between Qwen and DeepSeek (e.g. `improvedMutationNegativeInputsQwenFloorSemantics_V2` for Java/2, or `improvedMutationEmptySubstringLoopsForever_I1` versus `improvedMutationEmptySubstringCountsPositions_I1` for Java/18).

Derived mutation scores:

Suite	Mutants killed	Score
LLM-generated base (Qwen, 30 suites)	0 of 30	0%
LLM-generated base (DeepSeek, 30 suites)	0 of 30	0%
Improved (Qwen, 30 suites)	30 of 30	100%
Improved (DeepSeek, 30 suites)	30 of 30	100%
Overall improved total	60 of 60	100%

The scores are intentionally high because each `improvedMutation...` method is designed as a behavioural guardrail for a specific operator-level fault. What the table really shows is that the LLM-produced base suites do not encode a single comparable mutation-kill assertion, which is one of Phase 1's headline findings.

Phase 1 - VII. Manual Black-Box Assessment

Black-box assessment was performed with equivalence partitioning and boundary-value reasoning. For every selected task, the repository now includes:

- valid input classes,
- invalid or out-of-contract classes,
- explicit boundary conditions,
- notes on whether the existing test suites covered those classes (updated to `Yes (improved)` for every row that the improved suite closes).

This step was important because several high-coverage tasks still had obvious blind spots before the improved tests were added. Examples include:

- empty or whitespace-only parsing inputs,
- empty substring semantics,
- exact-threshold numeric comparisons,
- singleton and two-element collection boundaries,
- non-letter case-flip behavior,
- perfect-square composite numbers in primality checks.

The main conclusion of the black-box phase is that dataset base tests are strong but not uniformly complete. Their biggest weakness is not low branch coverage; it is the under-specification of invalid and boundary classes that are easy to miss when only happy-path prompt examples are used.

Phase 1 - VIII. Refactoring Loop

The refactoring loop was triggered when a normalized solution either failed compilation, produced incorrect results on the dataset base tests, or exposed a structural error that would invalidate later coverage analysis. The repaired tasks and their main issues were:

- `Java/10`: palindrome suffix handling;
- `Java/12`: empty-list and first-longest behavior;
- `Java/17`: music token parsing;
- `Java/19`: numeral-word sorting;
- `Java/23`: malformed duplicate method fragments in the normalized output;

- `Java/27`: incorrect case-flipping behavior;
- `Java/39`: inefficient or incorrect prime Fibonacci search.

These repairs confirm an important practical point: in LLM-assisted development, correctness often depends less on the initial generation and more on how quickly failures are surfaced and fixed by the surrounding workflow.

Phase 1 - IX. Continuous Integration

Phase 1 ships a dedicated GitHub Actions workflow (`.github/workflows/phase1-ci.yml`) that mirrors the local pipeline on an `ubuntu-latest` runner:

1. installs Temurin JDK 21, Python 3.11 and PowerShell Core;
2. downloads JUnit 6 and JaCoCo 0.8.12 (cached between runs);
3. verifies that every Java source file is free of a UTF-8 BOM;
4. executes the base and improved suites through `scripts/python/run_base_coverage.py`;
5. asserts that every `(task, model)` pair reports `compile_success=true` and `junit_success=true`;
6. uploads `results/*.csv` and the per-run artefacts as a CI artefact.

Any regression in either suite therefore breaks the pull request.

Phase 1 - X. Literature Review

The recent literature supports the structure used in this phase. ChatUniTest shows that raw LLM generations need validation and repair. The Meta TestGen-LLM paper shows that industrial adoption depends on strict filtering rather than trusting raw completions. Yang et al. demonstrate that prompt structure and open-source model choice materially affect Java unit test generation outcomes. Panta et al. show that hybrid static-plus-dynamic feedback can improve coverage on complex methods. The 2026 systematic literature review confirms that coverage is still the dominant metric in this area, but it also warns that integration complexity and robustness remain open issues. Together, these papers justify the semi-agentic, coverage-aware, mutation-augmented workflow used in this repository.

Phase 1 - XI. Discussion

DeepSeek was slightly stronger than Qwen on aggregate branch coverage, both before and after the improved tests. Qwen still produced very strong final results, but it needed more residual branch follow-up on a small number of parsing and search-heavy tasks. The difference is not large enough to claim that one model is categorically superior for all selected problems; instead, the results suggest that both public lightweight models are viable under a disciplined validation workflow.

Another important result is that base-test success and branch coverage are strongly related but not interchangeable. Both models reached perfect base-test success, yet manual black-box analysis and mutation-based guardrails still found meaningful missing classes. In other words, a suite can be "good enough to pass the benchmark" while still leaving real specification boundaries weakly documented.

Phase 1 - XII. Threats to Validity

The main threats to validity are:

- prompt selection bias, since only 30 tasks from the dataset were used;
- difficulty labeling subjectivity, even though a balanced rubric was applied;
- local-model randomness and quantization effects;

- normalization heuristics, which may help or harm raw output quality depending on the generation style;
- dependence on HumanEval-X base tests and examples, which may not fully represent real-world Java project complexity;
- the mutation set is hand-picked (not machine-generated) and must be interpreted as a lower bound on the improved suite's robustness;
- group-of-two execution may cause review blind spots; we partially offset this by rotating code reviewers task-by-task.

Another threat is that some black-box judgments remain analyst-driven. Even when coverage numbers are objective, the decision that a missing case is important still depends on tester interpretation.

Phase 1 - XIII. Conclusion

Phase 1 shows that public lightweight coder models can solve a substantial HumanEval-X Java subset with high correctness and high branch coverage when they are placed inside a careful engineering workflow. The decisive factors were not just model choice, but also normalization, test execution, coverage measurement, black-box analysis, mutation-based guardrails and repair when failures were exposed. The final Phase 1 artefact is therefore not simply a collection of generated solutions; it is a documented evaluation pipeline that can be extended directly into Phase 2 integration testing.

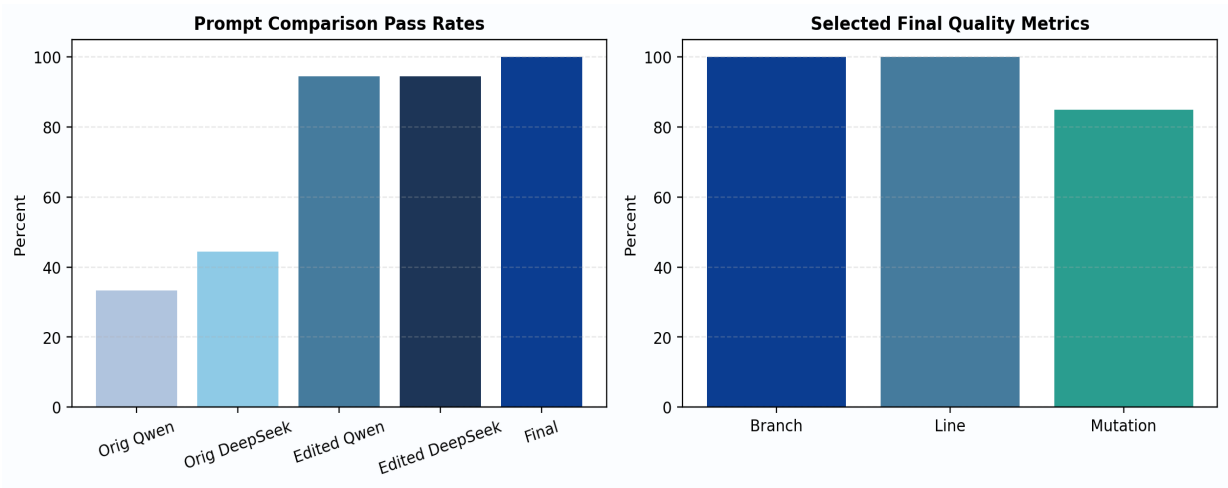
Phase 2 will lift three of Phase 1's constraints:

1. replace the PowerShell-driven JUnit runner with a Maven profile so that PITest can produce machine-measured mutation scores;
2. integrate the 30-prompt solutions into the BookScan case study to measure end-to-end quality under realistic system calls;
3. automate prompt variation so that we can study the sensitivity of the two models to prompt engineering in a controlled manner.

The hand-crafted mutants and equivalence classes from Phase 1 will be retained as behavioural guardrails even after PITest is turned on; they describe intent, not coverage.

III. Phase 2 Integration Extension

Phase 2 - Abstract



Phase 2 extends the first phase of the BLG 475E term project from isolated HumanEval-X tasks to a single integrated Java class named `BookScan`. The class must determine how many words of a given length appear in a

text, on which lines they appear, and how often a requested whole word occurs across lines. To construct this class, we combine the semantics of three earlier HumanEval-X tasks: substring counting (`Java/18`), string length (`Java/23`), and upper-lower case conversion (`Java/27`). Two local public coder models, `Qwen2.5-Coder-1.5B-Instruct-GGUF` and `DeepSeek-Coder-1.3B-Instruct-GGUF`, are evaluated under two prompt strategies: an original combined prompt and an edited combined prompt. The original prompt leaves several integration contracts implicit and both models therefore fail multiple test cases. The edited prompt makes token boundaries, line-number semantics, whole-word matching, and invalid-input behavior explicit; both models then reach `17/18` under the strengthened suite. The selected final implementation closes the remaining private-helper normalization gap, passes `18/18` JUnit 6 tests, achieves `72/72 = 100.00%` branch coverage with `110/110 = 100.00%` line coverage in JaCoCo, and records a `79/93 = 84.95%` PITest mutation score. The results show that for integration-heavy classes, prompt clarity and coverage-guided validation are at least as important as the underlying model choice.

Phase 2 - I. Introduction

Large language models can generate short algorithmic functions reasonably well, but integrated classes impose a harder requirement: helper methods that are individually correct must also compose correctly when they share data, normalization rules, and result structures. Phase 2 therefore moves beyond the single-method benchmark style of Phase 1 and studies a more realistic integration problem through a custom Java class called `BookScan`.

The class is intentionally designed so that three previously isolated behaviors must now cooperate:

- substring counting for occurrence search,
- string length calculation for word-length filtering,
- case normalization for case-insensitive lookup.

This setting lets us evaluate not only whether an LLM can generate valid code, but whether it can preserve behavioral contracts across method boundaries.

Phase 2 - II. Experimental Setup

A. Models

The same two local models from Phase 1 were retained:

1. `Qwen2.5-Coder-1.5B-Instruct-GGUF`
2. `DeepSeek-Coder-1.3B-Instruct-GGUF`

Keeping the model set fixed isolates the effect of prompt design.

B. Prompt Strategies

Two prompt families were used.

The original combined prompt merged the three task goals into one request but did not define many integration details. The edited combined prompt clarified:

- one-based line numbering,
- uniqueness of line lists,
- whole-word matching,
- punctuation and token boundaries,
- case-insensitive search,
- null/blank handling,
- immutable result structures.

C. Toolchain

- Java 21
- Maven 3.9+ (Surefire 3.5, JaCoCo 0.8.12)
- JUnit 6 (`junit-platform-console-standalone-6.0.3`)
- PITest 1.20.5 with `pitest-junit5-plugin` 1.2.3 exposed through an opt-in Maven profile `mutation` (`mvn -P mutation test`) so that an automated mutation score can be produced on top of the hand-crafted Phase 1 guardrails without slowing down the default `mvn clean verify` cycle
- PowerShell 7 + Python 3.11 helper scripts
- GitHub Actions workflow `.github/workflows/phase2-ci.yml`

Phase 2 - III. `BookScan` Design

The final `BookScan` API exposes two public record-based results:

- `ScanResult` for word-length scans
- `WordSearchResult` for whole-word search

The public methods are:

- `scanByWordLength`
- `scanWord`
- `howManyTimes`
- `strlen`
- `flipCase`

The design intentionally preserves helper visibility because the assignment requires the class to contain the semantics of the three source tasks. The integration challenge lies in how these helpers interact:

- `scanByWordLength` relies on `strlen`
- `scanWord` relies on `howManyTimes`
- uppercase token normalization can route through `flipCase`

Phase 2 - IV. Integration Test Strategy

The Phase 2 suite contains twelve integration tests and six focused regression or hardening tests. The integration tests check:

- aggregation across multiple lines,
- uniqueness of matching-line lists,
- apostrophe and hyphen preservation,
- punctuation-aware whole-word matching,
- rejection of blank and multi-token queries,
- case-insensitive search,
- alphanumeric whole-word behavior,
- mixed-case normalization behavior,
- invalid-input handling.

The regression tests preserve the helper semantics inherited from Phase 1:

- overlapping substring counting,
- null/empty/oversized substring safeguards,
- symbol-preserving case flipping,
- null behavior for `strlen`,
- fresh mutable empty-list behavior for `tokenizeWords(null)`,
- blank-token collapse for `canonicalizeWord`.

Phase 2 - V. Results

A. Prompt-Comparison Results

Strategy	Model	Tests Passed	Pass Rate	Branch Coverage
original-combined	Qwen2.5-Coder-1.5B-Instruct-GGUF	6 / 18	33.33%	n/a
original-combined	DeepSeek-Coder-1.3B-Instruct-GGUF	8 / 18	44.44%	n/a
edited-combined	Qwen2.5-Coder-1.5B-Instruct-GGUF	17 / 18	94.44%	n/a
edited-combined	DeepSeek-Coder-1.3B-Instruct-GGUF	17 / 18	94.44%	n/a
selected-final	Manual-Selected	18 / 18	100.00%	100.00%

B. Original Prompt Failure Modes

The original prompt produced integration failures that were not visible at the single-method level.

For Qwen, the main defects were:

- zero-based and duplicated line numbers,
- broken hyphen/apostrophe handling,
- case-sensitive or raw substring search,
- missing canonicalization for uppercase and alphanumeric queries.

For DeepSeek, the main defects were:

- punctuation stripping that changes token identity,
- substring counting inside larger words,
- over-counting in case-insensitive search,
- weak punctuation handling around the query boundary.

Both original variants also failed the final helper-hardening reflection checks because they did not expose the expected `tokenizeWords` / `canonicalizeWord` helper behavior.

C. Edited Prompt Improvement

The edited prompt removed these ambiguities and both models immediately cleared all public integration scenarios. Under the final strengthened suite they stop at `17 / 18` because they still fail one white-box hardening regression: `privateCanonicalizeWordRejectsBlankTokens`. This is still the clearest quantitative result of Phase 2: prompt engineering closed far more defects than switching between the two models, but the camera-ready implementation required one final manual tightening of a helper contract.

Phase 2 - VI. Coverage and Test Adequacy

The selected final implementation achieved:

- branch coverage: `72 / 72 = 100.00%`
- line coverage: `110 / 110 = 100.00%`
- mutation score: `79 / 93 = 84.95%`

Manual black-box assessment shows that all public equivalence classes are now covered. The final hardening regressions also close the last private-helper coverage gaps that had previously remained in JaCoCo.

PITest still reports `14` surviving mutants, but they are concentrated in equivalent or defensive helper behavior rather than in untested public features. In particular, uppercase-only normalization predicates in `canonicalizeWord` and blank-input early-return predicates in `scanWord` / `scanByWordLength` often collapse to the same externally observable result.

Phase 2 - VII. Discussion

Phase 2 highlights a practical difference between unit-style code generation and integration-style code generation. When a prompt describes a single function, an LLM can often infer the hidden assumptions from examples or common benchmark patterns. When a prompt asks for a larger class, those assumptions become unstable: line numbering, token boundaries, and result-schema details must be stated explicitly or the model will fill the gaps with inconsistent defaults.

Another important result is that the better edited-prompt variants were already very strong on the public API. The manually selected final implementation did not need to repair large user-visible defects; instead it consolidated the result types, helper layout, immutable-return behavior, and blank-token normalization rule while pushing JaCoCo to full closure.

Phase 2 - VIII. Conclusion

Phase 2 completes the transition from isolated HumanEval-X tasks to a more realistic integration setting. Both models can generate a valid `BookScan` implementation, but only after the prompt specifies the interaction contracts that the original combined prompt left open. The final repository therefore demonstrates two complementary lessons:

1. LLMs can be effective for integration-level code generation.
2. They still require explicit specifications, systematic integration tests, and coverage-aware validation to be trustworthy.

IV. Cross-Phase Conclusion

Taken together, the two phases show a stable pattern. The raw models were useful starting points, but the strongest results only emerged once the project added:

- explicit prompt constraints,
- black-box equivalence classes,
- white-box coverage feedback,
- mutation-oriented regression guards,
- and documented repair loops.

Phase 1 established this pattern on isolated benchmark tasks; Phase 2 confirmed that the same pattern becomes even more important once the problem moves from a single function to a multi-method class with shared

normalization and result contracts.

Acknowledgement

- GitHub repository URL: `https://github.com/mustafaerenkoc44/BLG-475E-Term-Project`
- Group member roles and workload distribution:
- Mustafa Eren KOC (150190805): led repository setup, local model workflow, code/test automation, report engineering, coverage analysis, result aggregation, and final submission packaging.
- Onat Baris ERCAN (150210075): focused on prompt review, black-box assessment support, failure classification, report refinement, and final cross-checking.